Kleinrock (2.92)

$$\frac{\partial}{\partial t} H(s,t) = H(s,t)\, Q(t)$$

$$\frac{d}{dt} \int_a^t f(t,u)\, du = f(t,t) + \int_a^t \frac{\partial}{\partial t} f(t,u)\, du$$

$$H_s(t) = e^{\int_s^t Q(u)\, du}$$

$$\frac{d}{dt} H_s(t) = \frac{d}{dt} e^{\int_s^t Q(u)\, du} = e^{\int_s^t Q(u)\, du} \cdot \frac{d}{dt} \int_s^t Q(u)\, du$$

$$= H_s(t) \cdot \left[Q(t) + \int_s^t \underbrace{\frac{\partial}{\partial t} Q(u)\, du}_{=0} \right]$$

$$\frac{\partial}{\partial t} H_s(t) = H_s(t)\, Q(t) \quad \Rightarrow \quad H_s(t) = e^{\int_s^t Q(u)\, du} \quad \text{is a solution} \;//$$

$$\pi(t) = \pi(0)\, H(0,t) = \pi(0)\, e^{\int_0^t Q(u)\, du}$$

$$\frac{d}{dt} \pi(t) = \underbrace{\pi_0(0)\, e^{\int_0^t Q(u)\, du}}_{} \cdot \left[Q(t) + \int_0^t \underbrace{\frac{\partial}{\partial t} Q(u)\, du}_{=0} \right]$$

$$\frac{d}{dt} \pi(t) = \pi(t)\, Q(t) \;//$$

$$\pi(t) = \pi(0)\, e^{Qt}$$

$$\frac{d}{dt} \pi(t) = \underbrace{\pi(0)\, e^{Qt}}_{\pi(t)} \cdot Q$$

$p_1 \quad p_2 \quad \cdots \quad p_N$

$$\pi^{(n)} = \begin{bmatrix} \pi_1^{(n)} & \pi_2^{(n)} & \cdots & \pi_N^{(n)} \end{bmatrix} = \begin{bmatrix} \pi_1^{(0)} & \pi_2^{(0)} & \cdots & \pi_N^{(0)} \end{bmatrix} \begin{bmatrix} p_{11}^{(n)} & & \\ p_{21}^{(n)} & P^n & \\ \vdots & & \\ p_{N1}^{(n)} & & \end{bmatrix}$$

$$\pi_1^{(n)} = \sum_{i=1}^{N} p_{i1}^{(n)} \pi_i^{(0)}$$

$$\pi_2^{(n)} = \sum_{i=1}^{N} p_{i2}^{(n)} \pi_i^{(0)}$$

$$\underbrace{\begin{bmatrix} \pi_1 & \pi_2 & \cdots & \pi_N \end{bmatrix}}_{} = \pi^{T(0)} p_1$$

$$\pi_1 = \pi_0^T p_1$$

$$\pi_2 = \pi_0^T p_2$$

$$\pi_3 = \pi_0^T p_N$$

$$\begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix} = \begin{bmatrix} \pi_0 & 0 & 0 \\ 0 & \pi_0 & 0 \\ 0 & 0 & \pi_0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

3×1 $\quad 3 \times 9 \quad 9 \times 1$